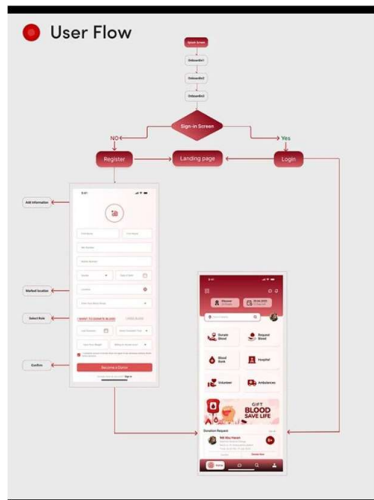


PROJECT NAME : BLOOD BANK AND RARE DONAR NETWORK

1.Overall approach



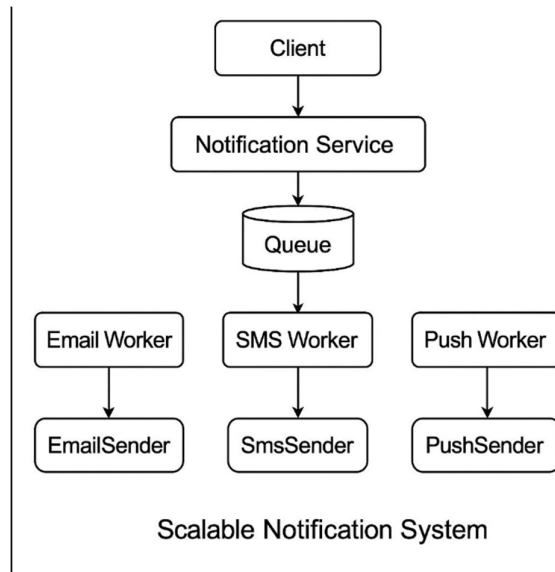
The system addresses the major problem of disconnected blood banks and difficulty in finding rare donors by implementing a centralized, real-time digital platform.

✓ Core Strategy:

- Replace manual & isolated systems with a connected network
- Enable instant donor discovery
- Ensure fast emergency response

“Right blood → Right person → At the right time”

2. System Architecture Overview



✖ 1. Frontend (Presentation Layer)

- Mobile/Web App (User Interface)
- Used by:
 - Donors
 - Recipients
 - Blood Banks

Features:

- Register/Login
- Search blood by group & location
- Request blood instantly
- Track request status

⚙ 2. Backend (Business Logic Layer)

- Core processing unit of the system

Responsibilities:

- Validate user data

- Match donors with recipients
- Handle emergency requests
- Manage request lifecycle:

Requested → Accepted → Donated → Completed

3. Database (Data Layer)

Stores:

- Donor details
- Blood groups
- Location data
- Request history

 Ensures:

- Fast retrieval
- Data consistency
- Real-time updates

3. Key Components

1. Donor Matching Engine

- Matches based on:
 - Blood group
 - Location (nearest donor)
 - Availability
-

2. Emergency Alert System

- Sends notifications via:
 - SMS / Push Notifications

- Alerts nearby donors instantly

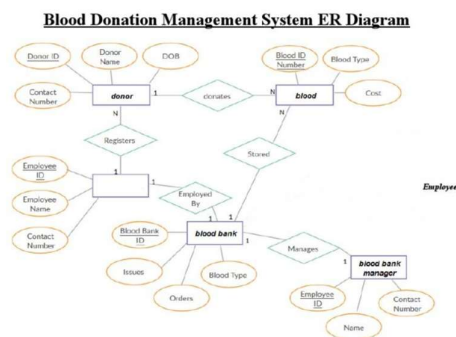
3. Blood Bank Integration Module

- Connects multiple blood banks
- Shares:
 - Blood availability
 - Stock levels

4. Admin Dashboard

- Monitor:
 - Total requests
 - Active donors
 - Emergency case

4. Data Flow Explanation



Step-by-Step Flow:

1. User Request

- Recipient searches for blood
- Inputs: Blood group + location

2. Processing

- Backend checks:
 - Nearby donors
 - Blood bank availability

3. Matching

- Best donors are selected

4. Notification

- Alerts sent to donors

5. Response

- Donor accepts request

6. Completion

- Donation done
- Status updated

5. Design Decisions

Web-Based Platform

- Accessible anywhere (mobile + laptop)
 - No installation required
-

Location-Based Matching

- Uses real-time location
 - Ensures faster response
-

Priority System

- Emergency requests highlighted
 - Faster processing for critical cases
-

User-Friendly UI

- Simple forms
- Clear status indicators
- Easy navigation